

Elaine Hong

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EDUCATION

Cornell University, College of Engineering

B.S. in Computer Science and Operations Research and Information Engineering

Ithaca, NY

Expected May 2028

EXPERIENCE

Software Engineer Intern

June 2026 – August 2026

FinMD

Remote

- Developed production underwriting and portfolio-management workflows using **Next.js**, **TypeScript**, **Supabase**, and **PostgreSQL**
- Built tenant-authenticated **APIs and PostgreSQL RPCs** to prefill **10+** fields across product workflows, reducing duplicate borrower data entry and increased form completion speed by **40%**
- Implemented server-side pagination, filtering, and sorting for **1,000+** tenant records, limiting responses to 25 records per page and increased loading speed by **60%**

Machine Learning Engineer Intern

June 2026 – August 2026

Anote AI

New York, NY

- Built **Python** and **LangChain** evaluation frameworks to test whether common RAG enhancements improved financial-document question answering across **133** FinanceBench questions.
- Automated controlled experiments across semantic and recursive chunking, query expansion, retrieval, and cross-encoder reranking, evaluating answer quality, numeric accuracy, latency, and model cost to assess gains over the baseline
- Developed **React** and **Redux** workflows for AI campaign configuration, prospect targeting, and real-time outreach tracking, while implementing OAuth integrations for email and Slack with token lifecycle and API rate-limit handling

Research Assistant

August 2025 – June 2026

Cornell Engineering Undergraduate Research

Ithaca, NY

- Built **Python** pipelines to compare **LLM** move evaluations with **Stockfish** scores across **1,000+** chess positions, supporting analysis of strategic reasoning errors
- Designed and evaluated **20+** prompting strategies, measuring move quality and error rates across **200+** chess games

PROJECTS

Algorithmic Trading System | *Python, FastAPI, MySQL, Redis, Docker, WebSockets*

- Built a modular trading backend with strategy execution, order management, and real-time market-data ingestion through **WebSockets** feeds
- Developed **FastAPI REST** and **WebSockets** endpoints for strategy control and portfolio updates, using **Redis** to cache positions and portfolio state
- Containerized backend services with **Docker** and designed **MySQL** persistence for trades, orders, and strategy results

IMC Prosperity 4 | *Python, Pandas, NumPy, Algorithmic Trading*

- Developed and backtested market-making strategies using simulated order-book data across multiple products
- Ranked in the top **10%** globally by adapting pricing and risk strategies across changing market conditions

Bear Escape — BigRed//Hacks Winner | *React, TypeScript, Python, Gemini API, OpenCV, SpaceTimeDB*

- Built an AI-powered rhythm game with **React**, **TypeScript**, **Python**, **Google Gemini**, **OpenCV** in **36** hours
- Won **Best Use of Gemini** and **Best UI** at an event with **200+** participants for technical execution and product design

LEADERSHIP & ACTIVITIES

NCAA Division I Varsity Fencing Athlete

August 2024 – Present

Cornell University

Ithaca, NY

- Balance **20+** hours per week of Division I training, travel, and competition with Cornell engineering coursework
- Compete as part of a **25+** member team, demonstrating discipline, adaptability, and performance under pressure

Conference Director

April 2026 – Present

Society of Women Engineers at Cornell

Ithaca, NY

- Lead planning for an annual engineering conference, managing an **\$8,000** budget and logistics for **20+** attendees
- Coordinate sponsors, reimbursements, and travel across multiple workstreams to meet deadlines and budget targets

TECHNICAL SKILLS

Languages: Python, Java, C++, TypeScript, JavaScript, SQL, C

Frameworks & Libraries: FastAPI, React, Next.js, Redux, LangChain, Pandas, NumPy, OpenCV

Databases & Backend: PostgreSQL, MySQL, Redis, Supabase, Firebase, SpaceTimeDB

Infrastructure & APIs: AWS EC2, Docker, REST APIs, WebSockets, OAuth, Git, GitHub